

LabLens UI Redesign Report

Columbia Claude Hackathon — April 2026
Competitive Analysis & Improvement Strategy

Project Architecture Summary

LabLens is an AI-powered research alignment engine that helps graduate students find matching professors.

- **Frontend:** Next.js 14, Tailwind CSS, Framer Motion, TypeScript (4 pages)
- **Backend:** Python FastAPI + sentence-transformers (ML) + Claude Sonnet + Linkup API
- **External APIs:** arXiv (papers), NSF (grants), Linkup (live web search)
- **Novel Feature:** Differentially private three-tier matching with Gaussian noise mechanism
- **Pipeline:** 6-stage (Intent → Search → Verify → Embed → Rank → Synthesize)

Current UI Problems

Issue	Why It Hurts
Playfair Display (serif) headings	Reads “luxury editorial,” not “intelligent research tool”
4 competing accent colors	Amber, teal, coral, purple all fight for attention — no visual hierarchy
Animated dot-grid background	Distracting pulsing animation competes with content
Emoji badges	Brain, lightning, chain emojis look unprofessional in academic context
Glass-morphism too transparent	Cards at <code>rgba(17,24,39,0.6)</code> look washed out against <code>#0B0E1A</code>
JetBrains Mono overused	Monospace on badges, buttons, helper text — should be data-only

Recommended UI References

1. Perplexity AI — Search Interface Gold Standard

- **URL:** <https://perplexity.ai>
- Single accent color (cyan `#20b8cd`), narrow content width (`max-w-[42rem]`)
- Pipeline transparency: “Considering 8 sources... Researched and summarized”
- FK Grotesk font, clean focus rings, predicted follow-up questions
- **Apply to LabLens:** Constrain search input width, show specific data in loading steps (e.g., “Analyzing 847 papers from Prof. Smith...”), reduce to one accent color

2. Consensus.app — Academic Search Results Cards

- **URL:** <https://consensus.app>
- Paper title at TOP of each card, AI-extracted answer below in grey box
- “Study Snapshot” shows key data points at a glance
- “Consensus Meter” visualization for topic agreement
- **Apply to LabLens:** Lead ProfessorCard with name (not score circle), add “Research Snapshot” section, replace alignment bar with compact badge

3. Elicit — Structured Research Data UI

- **URL:** <https://elicit.com>
- Card-based interface with structured table columns (Findings, Methods, Limitations)
- Click any data point to reveal exact source quote — builds trust
- “Add Column” for custom AI-extracted questions across all results
- **Apply to LabLens:** Add table/list view toggle, “Why this match?” expandable section linking alignment scores to specific papers

4. Linear App — Dark Mode Developer Tool Standard

- **URL:** <https://linear.app>
- **Analysis:** <https://frontend.horse/articles/the-linear-look/>
- THE defining aesthetic of 2024–2025 dark-mode tools
- Near-black surfaces (#111–#1a1a1a), super thin 1px borders at 6% opacity
- Colorful blurry glows as decoration, bento box grid layouts
- Sans-serif typography is “an absolute law”
- **Apply to LabLens:** Make dot grid static, add blurred color orbs (`filter: blur(100px)`), remove box-shadow glows, use border-color hover transitions

5. shadcn/ui — Component Foundation

- **URL:** <https://ui.shadcn.com> (112k+ GitHub stars)
- The de facto standard for Next.js + Tailwind in 2025–2026
- Cmd+K Command palette for global search, built on Radix UI (accessible)
- Key components: Command, Card, Badge, Dialog, Tabs, Avatar, Skeleton, Sheet
- **Apply to LabLens:** Install shadcn/ui, use Command component for search bar, Sheet for slide-in professor details, Skeleton for loading

6. Aceternity UI — Animation Layer

- **URL:** <https://ui.aceternity.com>
- 200+ animated components, dark-mode-first, Framer Motion + Tailwind
- Key effects: Aurora Background, Lamp Effect, Text Generate Effect, 3D Card Effect
- Copy-paste integration — each component is self-contained
- **Apply to LabLens:** Aurora/Lamp Background on homepage hero, Text Generate Effect for tagline, 3D Card hover on ProfessorCards

7. Magic UI — Marketing Polish

- **URL:** <https://magicui.design>

- Built on shadcn/ui, composable together
- Key components: Bento Grid, Magic Card (cursor spotlight), Animated Gradient Text, Retro Grid
- Uses Geist + Geist Mono fonts (Vercel's typefaces)
- **Apply to LabLens:** Replace dot grid with Retro Grid, use Magic Card for professor cards, Bento Grid layout with featured #1 professor in double-wide card

8. Vercel Dashboard Template — Production Reference

- **URL:** <https://vercel.com/templates/next.js/next-js-and-shadcn-ui-admin-dashboard>
- Official Vercel template — what Next.js apps “should” look like
- Sticky sidebar, Command palette search, card-based metrics, responsive
- **Apply to LabLens:** Add metric cards at top of results (“12 professors found · 847 papers analyzed · 78% avg alignment”)

9. v0.dev by Vercel — Instant UI Generation

- **URL:** <https://v0.dev>
- Describe any UI in natural language → production React + Tailwind + shadcn code
- Free tier: 200 monthly generations
- **Apply to LabLens:** Generate alternate professor card designs, hero sections, results layouts instantly

10. Semantic Scholar — Academic Tool Reference

- **URL:** <https://semanticscholar.org>
- AI-generated TLDRs on every paper card, citation graph visualization
- “Skimming Highlights” labels: Goal, Method, Result with colored overlays
- **Apply to LabLens:** Add one-line TLDR per professor card, Goal/Method/Result labeling pattern for profile sections

11. TailAdmin Next.js — Free Dashboard Template

- **URL:** <https://github.com/TailAdmin/free-nextjs-admin-dashboard>
- Free, MIT-licensed, 7 dashboard variants, 500+ components
- Next.js App Router + TypeScript, full dark mode
- **Apply to LabLens:** Fork and restyle if rebuilding — has every page type needed

12. Linear Look Theme — Exact Color Specification

- **URL:** <https://github.com/Arroz-11/ComfyUI-Linear-Theme>
- Codified the exact Linear/Vercel/Raycast aesthetic into a reusable theme
- Formula: dark background + single indigo accent + thin borders + no box shadows
- **Apply to LabLens:** Consider shifting from amber to indigo (#6366F1) — amber reads as “warning,” indigo reads as “intelligent, trustworthy, academic”

Consolidated Improvement Prompt

The following prompt can be used with v0.dev, Claude, or Cursor to generate an improved UI:

Redesign the LabLens AI research matching tool using these principles:

Typography: Replace Playfair Display with Inter or Geist Sans. Use Geist Mono for data/scores only. All headings sans-serif.

Colors: Dark background #09090B (zinc-950), surfaces #18181B (zinc-900), ONE primary accent (indigo #6366F1 or amber #F59E0B), ONE secondary accent (cyan #22D3EE). Remove coral and purple from active use.

Hero page: Aceternity-style Aurora Background or Lamp Effect behind the title. Text Generate Effect on the tagline “Find your research home.” Remove dot-grid animation. Add 2–3 blurred gradient orbs (`filter: blur(100px), 10% opacity`). Replace emoji badges with small colored dots + text.

Search: shadcn/ui Command component (Cmd+K). Constrain to `max-w-2xl`. Single glass card, clean focus ring.

Results page: Summary metrics bar at top (“12 professors found | 847 papers analyzed | Avg. alignment: 78%”). Bento Grid layout — #1 professor gets double-wide card. Each card: Name first (large), department, match score as small badge top-right, one-line TLDR, compact stats row. Magic UI cursor-spotlight hover effect.

Professor detail: Slide-in Sheet panel instead of full page navigation. Keep 6 tabs using shadcn Tabs. Show “Why this match?” with specific paper citations.

Loading: Specific data in pipeline steps (“Analyzing 12 papers from Prof. Smith...”). Use shadcn Skeleton components.

Borders: `rgba(255,255,255,0.06)`, 1px only. No box-shadow glows — use border-color changes on hover. Rounded corners: 12px cards, 8px buttons, 6px badges.

Animations: Framer Motion staggered entrance (200–400ms). Layout animations on sort/filter. No pulsing backgrounds.

Recommended Palette & Font Replacement

Current	Suggested	Role
#0B0E1A	#09090B	Background (zinc-950)
#111827	#18181B	Surface (zinc-900)
#1A2035	#27272A	Tertiary (zinc-800)
#F59E0B	#6366F1	Primary accent (indigo-500)
#14B8A6	#22D3EE	Secondary accent (cyan-400)
#8B5CF6	---	Remove or use very sparingly
#F87171	---	Remove

Current Font	Suggested Font	Role
Playfair Display (serif)	Inter or Geist Sans	Display headings
DM Sans	Keep or switch to Inter	Body text
JetBrains Mono	Geist Mono	Data & scores only

Priority Action Plan

Tonight (Biggest Visual Impact, Least Code)

1. Replace Playfair Display with Inter/Geist Sans (one config change)
2. Kill dot-grid animation (remove `animation` property)
3. Add blurred gradient orbs to hero (2–3 absolutely positioned divs)
4. Drop to 2 accent colors (amber primary, teal secondary)
5. Replace emoji badges with colored dots + text

Tomorrow (Moderate Effort, High Impact)

6. Install shadcn/ui, replace search with Command component (Cmd+K)
7. Add Aceternity Aurora Background or Lamp Effect to homepage
8. Restructure ProfessorCard: name first, score as small badge, one-line TLDR
9. Add summary metrics bar at top of results page
10. Use v0.dev to generate 2–3 alternate card designs

If Time Permits

11. Add shadcn Sheet for slide-in professor details (faster demo flow)
12. Add Framer Motion `layout` animations on sort/filter
13. Implement Bento Grid with double-wide card for top professor